

INTERVENTIONAL PAIN & SPINE MEDICINE SECTION

Simplified Universal Grading of Lumbar Spine MRI Degenerative Findings: Inter-Reader Agreement of Non-Radiologist Spine Experts

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Abstract

Objective. 1) To describe a simplified multidisciplinary grading system for the most clinically relevant lumbar spine degenerative changes. 2) To measure the inter-reader variability among non-radiologist spine experts in their use of the classification system for interpretation of a consecutive series of lumbar spine magnetic resonance imaging (MRI) examinations. **Methods.** ATS multidisciplinary and collaborative standardized grading of spinal stenosis, foraminal stenosis, lateral recess stenosis, and facet arthropathy was developed. Our institution's picture archiving and communication system was searched for 50 consecutive patients who underwent non-contrast MRI of the lumbar spine for chronic back pain, radiculopathy, or symptoms of spinal stenosis. Three fellowship-trained spine subspecialists from neurosurgery, orthopedic surgery, and physiatry interpreted the 50 exams using the classification at the L4–L5 and L5–S1 levels. Inter-reader agreement was assessed with Cohen's kappa coefficient. **Results.** For spinal stenosis, the readers demonstrated substantial agreement ($\kappa = 0.702$). For foraminal stenosis and facet arthropathy, the three readers demonstrated moderate agreement ($\kappa = 0.544$, and 0.557 , respectively). For lateral recess stenosis, there was fair agreement ($\kappa = 0.323$). **Conclusions.** A simplified universal grading system of lumbar spine MRI degenerative findings is newly described. Use of this multidisciplinary grading system in the assessment of clinically relevant degenerative changes revealed moderate to substantial agreement among non-radiologist spine physicians. This standardized grading system could serve as a foundation for interdisciplinary communication.

Key Words: Spine; Low Back Pain; Lumbar; Degenerative Disc Disease; Magnetic Resonance Imaging; Spinal Stenosis; Facet Joint

Introduction

Low back pain, radiculopathy, and neurogenic claudication together comprise one of the most common reasons for seeking clinical care [1–3] and have a tremendous societal burden [4, 5]. Imaging patients with low back pain continues to be a challenging issue. Prior work has shown that imaging often identifies abnormalities in asymptomatic patients [6–10], and imaging findings may not predict the future development of back pain [11]. Despite

these issues, as recommended by the guidelines of the American College of Physicians and the American Pain Society, magnetic resonance imaging (MRI) has a key role for patients with persistent low back and leg pain who are potential candidates for treatment procedures, including injections and surgery [12]. The fundamental principal of interventional treatment of lumbar spine disorders is to target the specific site(s) of pathology that correlate with clinical symptoms, and MRI is the primary